

Ryan Ptomey

Full-Stack Software Engineer (Python / JavaScript)

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Professional Summary

Full-stack software engineer with 4+ years of experience building data-driven web applications, automation tools, and scalable cloud infrastructure, supported by a deep background in scaling technical operations and digital infrastructure. Skilled at connecting APIs, cloud services, and event-driven architectures to streamline workflows and enable new business capabilities. Strong foundation in backend development, with deep production expertise in Python, JavaScript, and Google Cloud Platform.

Skills

Languages: Python, JavaScript, SQL

Frameworks & Web: Django, Flask, Streamlit, Astro, HTML, CSS, Bootstrap, REST APIs, Google Tag Manager

Cloud & DevOps: Google Cloud Platform (Cloud Run, Cloud Run Functions, BigQuery, Pub/Sub, Dataform, Workflows, Cloud Storage, Firestore), Terraform, Docker, CI/CD

Data & Tools: PostgreSQL, MySQL, MS SQL Server, Git (GitHub, Bitbucket), VS Code, Zsh, Copilot, Gemini

Experience

Data Engineer 2 (Software & Infrastructure Focus)

Oct 2025 - Present

Adswerve

- **Architected an event-driven GCP orchestration pipeline** (Python, Cloud Functions, Pub/Sub, Firestore) to automatically remediate failing SSL certificates for 2,500+ domains, integrating BigQuery log routing sinks to monitor real-time infrastructure and traffic health.
- **Recovered and refactored legacy Terraform configurations** from a raw state file, successfully restoring automated SSL certificate management for thousands of domains and eliminating manual deployment overhead.
- **Co-developed a full-stack media analytics application** (Streamlit, Python, GCP) integrating Google's open-source ABCDs Detector to ingest MP4 files, manage Cloud Storage uploads, and evaluate 70+ video features.
- **Architected relational BigQuery schemas** and engineered a dynamic UI feature-weighting system with advanced search/filter mechanics, enabling cross-functional teams to seamlessly deploy downstream campaign forecasting tools.
- **Prototyped generative AI features** in Streamlit to provide narrative interpretations of dynamic visualizations and accelerate internal data analysis workflows.
- **Integrated robust REST APIs** for secure, high-throughput data delivery to external platforms including Facebook and Campaign Manager 360.

Data Engineer 1 (Applications & Automation)

Jul 2022 - Oct 2025

Adswerve

- **Developed full-stack web automation tools**, including a Django application for dynamic IP range management and an automated video validation app comparing uploaded files to technical specifications.
- **Created server-side tracking infrastructure** on GCP, enabling secure, containerized, and scalable event forwarding for enterprise-level web applications.
- **Maintained and enhanced open-source SDKs**, contributing core Python functionality to the GA4 Measurement Protocol library to support advanced e-commerce data models.
- **Authored reusable JavaScript automation scripts** utilizing the Intersection Observer API to streamline enterprise video player tracking across client web properties.

Senior Digital Analyst

Aug 2019 - Jul 2022

Adswerve

- **Implemented and maintained enterprise codebases** across 30+ production websites using HTML, CSS, and JavaScript.
- **Engineered data collection architectures** for complex single-page applications, significantly improving event accuracy and reducing data loss.
- **Created templated architecture containers** and reusable code libraries, reducing engineering deployment times by 40%.
- **Delivered technical training** to developers and analysts on analytics instrumentation, API integrations, and system debugging workflows.
- **Collaborated with engineering teams** to programmatically integrate Enhanced Ecommerce tracking frameworks across web and mobile platforms.

Manager of Web Analytics

Apr 2018 – Aug 2019

Alight Analytics

- **Managed a engineering-adjacent team** of specialists delivering custom data pipelines and specialized reporting software for B2B and B2C clients.
- **Developed and scaled custom JavaScript tracking scripts** and business intelligence dashboards in Tableau and Data Studio.
- **Partnered closely with clients** to architect cross-platform data collection strategies aligned to automated business goals.
- **Automated recurring reporting engines** and platform onboarding processes, directly improving data delivery speeds and operational consistency.

Marketing Insight Analyst

Feb 2016 – Apr 2018

Alight Analytics

- **Deployed tag management scripts** across enterprise client sites to monitor web infrastructure and marketing application performance.
- **Built custom JavaScript utilities** and interactive Tableau dashboards to validate and audit pipeline data accuracy.
- **Partnered with development teams** to systematically troubleshoot, patch, and optimize application event logging.

Account Manager → Sr. Account Manager → Account & Product Coordinator

May 2009 – Dec 2015

Adknowledge

- **Transitioned to technical operations** within a high-volume advertising ecosystem, introducing custom process automation scripts and programmatic data reporting for engineering teams.

Education & Certifications

- **CS50w: Web Programming with Python and JavaScript** : Harvard University (2021)
- **CS50x: Computer Science Fundamentals** : Harvard University (2021)
- **M.A. Marketing Communications** : University of Kansas
- **B.A. English** : Loyola University Chicago

Open Source & Technical Contributions

- **Creator, GCP SFTP Server for GA4 Data Import:** Developed and published an open-source Terraform project automating secure Compute Engine and Cloud Storage provisioning. GitHub → [adswerve/gcp-sftp-server-for-ga4-data-import-terraform](#)
- **Core Contributor, GA4 Measurement Protocol (Python SDK):** Enhanced server-side event collection capabilities by adding support for advanced e-commerce data models and parameters. GitHub → [adswerve/GA4-Measurement-Protocol-Python](#)